

- Q.26 What is the difference between circuit and sub-circuit?
- Q.27 Draw the circuit of stair case installation.
- Q.28 Differentiate between primary and secondary distribution system.
- Q.29 What is the difference between primary and secondary transmission system.
- Q.30 Write various applications of Single phase motors and three phase motors.
- Q.31 Define star-delta starter with diagram.
- Q.32 What are the advantages of H.R.C Fuses?
- Q.33 Write various precautions against electric shock.
- Q.34 What is Zener diode? Write its application.
- Q.35 Define Transistor.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Draw the key diagram of Transmission system and explain every components in details.
- Q.37 Explain plate earthing details with diagram.
- Q.38 Explain various applications of electricity.

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1st Sem / Mech. Engg (MSIL)
Subject:- Basics of Electrical and Electronics Engg.

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 A direct current which has constant magnitude is called
- a) Steady DC b) Pulsating DC
- c) Alternating DC d) None of the above
- Q.2 In India, the frequency of AC is
- a) 30Hz b) 40Hz
- c) 50Hz d) 60Hz
- Q.3 The Unit of current is
- a) Watt b) Volts
- c) Ohm d) Ampere
- Q.4 The Instrument used to measure energy Consumed in an electric circuit is called
- a) Wattmeter b) Energy Meter
- c) Voltmeter d) Ammeter

- Q.5 The winding connected to source of AC is known as
 a) Secondary winding b) Transformer
 c) Primary winding d) Core winding
- Q.6 Which oil is used in transformer for cooling
 a) Mineral oil b) Soluble oil
 c) Synthetic fluid d) None of the above
- Q.7 The Colour of Earth wire is
 a) Red b) Black
 c) Green d) Yellow
- Q.8 Which supply is mostly used in industries for running of High motors.
 a) Single phase b) Four phase
 c) Two phase d) Three phase
- Q.9 A material having 3 electrons in its valence orbit is known as
 a) P-type Semi conductor
 b) N-type Semiconductor
 c) PNP Transistor
 d) NPN Transistor
- Q.10 The Section on one side of the transistor that supplies a large number of majority Carriers are called
 a) Base b) Collector
 c) Diode d) Emitter

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 The frequency of AC in India.
- Q.12 What is Direct current?
- Q.13 What is the unit of electrical energy?
- Q.14 Define electric current.
- Q.15 Define wattmeter.
- Q.16 What is Secondary winding?
- Q.17 What is Neutral wire?
- Q.18 Write full form of MCB.
- Q.19 What is Motor.
- Q.20 What is Earthing?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What are the advantages of DC over AC?
- Q.22 What is the difference between AC and DC?
- Q.23 Define frequency. Derive its unit? Draw a circuit to measure its value.
- Q.24 Explain step-down transformer.
- Q.25 State Faraday's law of electromagnetic induction.